TEST REPORT

Report No……..……..:	HST202107-2491
Sample Description….. :	SOLAR CHARGER
Model/Type……..:	2045
Client ……..…. :	YIWU MOJIA TRADING CO.,LTD
Assessment Category..:	Entrusted

Guangdong Huesent Testing & Inspection Technology Co., Ltd

TEST REPORT

Report Reference No ……:		
Tested By (+ Signature). ……:	Terry	Tested By :Terry
Reviewed By (+ Signature) ……:	Sam	Inspected By: Sam
Approved By (+ Signature) ……:	Louis	Approved By: Louis
Date Of Issue ……:	2026.02.17	
Client		
Applicant ……:	YIWU MOJIA TRADING CO.,LTD	
Address …… :	CHINA	
Manufacturer……:	YIWU MOJIA TRADING CO.,LTD	
Address……..……..……:	CHINA	
Testing Laboratory		
Name……:	Guangdong Huesent Testing & Inspection Technology Co.,Ltd.	
Address ……:	Room 401,4F,Section 2,No.600,Guangshan No.2 Road,Tianhe District,Guangzhou City,Guangdong	
Laboratory Qualification ……:	Laboratory has been accredited by CNAS (China National Accreditation Service for Conformity Assessment) and CMA (China Metrology Accreditation), The CNAS registration number is L2885.The CMA registration number is 2014191614Z.	
Test Specification		
Standard ……:	IEC 60335-1;IEC 60335-2-29-2016 Household and similar electrical appliances -- Safety -- Part 2-29: Particular requirements for battery chargers	
Sample Received Date……:	2026.02.09	
Test Duration…...……:	2026.02.09 ~ 2026.02.17	
Conformity ……:	☒ Yes ☐ No	
Test Item		
Description ……:	SOLAR CHARGER	
Trademark ……:	SUNAFRICA	
Model And/Or Type Reference ……:	2045	
Batch No……:	SUM202639-4	
Ratings……..……..……:	Input:DC 12V output:4-6V=400mA	
Remark: The sample conforms to the inspected items, and the inspected sample is qualified.		

ITEMS FOR ATTENTION

1. It would be invalid test report without specific stamp for test institute or the authority.

2. It would be invalid duplicated report without specific stamp for test institute or the authority.

3. It would be invalid test report without all the signatures of compilation, reviewer and approver.

4. It would be invalid test report, if there is any scrawl in the test report without official authorization.

5. Any disputes about the report must be submitted for test institute within 15 days from the day when the report is received, otherwise that would be invalid out of expiry.

6. Generally, the responsible is only for the samples in entrusted test.

Remark: Possible test case verdicts:
Test case does not apply to the test object……:N(.A.)
Test item does meet the requirement……:P(ass)
Test item does not meet the requirement……:F(ail)

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IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
4	General conditions for the tests		P
	Tests performed according to Cl. 4, e.g. nature of supply, sequence of testing, etc.		P
4.2	If the tests of 24.1.3 are carried out, three switches or three additional appliances are needed		N
4.8.1	For a.c. only, tested at rated frequency	Rated frequency:/	N
	For a.c./d.c., tested at the most unfavourable supply		N
	Appliances not marked with rated frequency, tested with 50 Hz		N
	Appliances for a.c. with frequency range of 50 to 60 Hz, tested with 50 or 60 Hz, the most unfavourable		N
5	General conditions for the tests		P
5.8.2	If an appliance is marked for use at two different rated voltages, and the instructions indicate that it is suitable for use at two different ranges of voltages, the appliance is assumed to have two rated rated voltage ranges.		P
6	Classification		P
6.1(60335-2-29)	Hairdryers, curling irons, curling combs, facial saunas and other steam-producing or spray-producing appliances shall be class II or class III. However, fixed hairdryers intended to be permanently connected to fixed wiring, helmet-type hairdryers for hairdressers and steam-producing or spray-producing appliances for hairdressers may be class I	N/A	P
6.2(60335-2-29)	Hand dryers shall be at least IPX1.		N
	Curling rollers of permanent-wave appliances shall be at least IPX4.	Not such appliance	N
7	Marking		P
7.1	Rated voltage or voltage range (V) .....:	Input:DC 12V	P
	Nature of supply		P
	Rated frequency or frequency range (Hz) .....:		P
	Rated power input or rated current	400mA	P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Manufacturer's or responsible vendor's name, trademark or identification mark	YIWU MOJIA TRADING CO.,LTD	P
	Model or type reference	100A	P
	Symbol for Class II		P
	IP number		N
7.1(60335-2-29)	Portable hairdryers, curling irons and similar appliances shall be marked with symbol 5582 of IEC 60417-1 combined with the prohibition sign of ISO 3864, except for the specified colours, or with the substance of the following:		P
	Warning: Do not use this appliance near water.		P
7.2	Warning for stationary appliances	Not stationary appliances	N
	Warning placed in vicinity of terminal cover		N
7.3	Range of rated values correctly marked		P
7.4	Voltage setting clearly discernible		P
7.5	Marking of rated input for each rated voltage		N
	Marking for upper and lower limits of rated input		N
7.6	Correct symbols used	Refer to marking label	P
	Additional symbols		P
7.6(60335-2-29)	[symbol 5582 of IEC 60417-1] Suitable for use in a bath or shower		N
7.7	Correct connection diagram, fixed to the appliance		P
7.8	Not for type Z attachment:		P
	- marking of terminals for the neutral conductor (N)		N
	- marking of earthing terminals		P
	- marking not placed on removable parts		N
	- marking of terminal for single-pole protective device		N
7.9	Marking or placing of switches which may cause a hazard		P
7.10	Indications of switches and regulating devices by use of figures, letters or other		P
	The figure 0 indicates only OFF position, unless no confusion with the OFF position	ON, OFF	P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
7.11	Indication for direction of adjustment of controls		P
7.12	Instructions for safe use provided	Safe in user manual	P
	Appliances incorporating batteries which contain materials hazardous to the environment: statement in the instructions how to remove, scrap and dispose of the battery safely	No batteries	N
	Statement in the instructions that the appliance must be disconnected from the supply		P
(60335-2-29)	WARNING: Do not use this appliance near bathtubs, showers, basins or other vessels containing water.	Warning in user manual and figure in marking label	P
7.12.1	Sufficient details for installation or maintenance supplied	Sufficient details in user manual	P
7.12.2	Means for disconnection with contact separation at least 3 mm		P
	Stationary appliances without power supply cord: hint in instruction about disconnection from mains		N
	Stationary appliance with supply cord and plug: statement in the instructions that the appliance is so positioned that the plug is accessible		N
7.12.3	Insulation in contact with parts exceeding 50 K; instruction		N
7.12.4	Information with regard to building-in:		N
	- dimensions of space		N
	- dimensions and position of support		N
	- ventilation openings		N
	- connection/interconnection plug accessible		N
7.12.5	Replacement cord, type X attachment		N
	Replacement cord, type Y attachment	Statement in user manual	P
	Replacement cord, type Z attachment		N
7.13	Instructions and other texts in official language	In English	P
7.14	Marking easily legible and durable	The marking is legible and durable	P
(60335-2-29)	The diameter of the circle superimposed on symbol 5582 of IEC 60417-1 shall be at least 10 mm.		N
7.15	Marking on a main part	On enclosure	P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Marking clearly discernible from outside	The marking is clear on enclosure	P
	Stationary appliance: name or trademark and model or type reference visible after installation		N
	Indication for switches and controls in vicinity of components; not on removable parts if misleading	Indication for switch: ON/OFF	P
7.16	Marking of a possible replaceable thermal link or fuse link clearly visible with regard to replacing the link		N
8	Protection against accessibility to live parts		P
8.1	Adequate protection against accidental contact with live parts		P
8.1.1	All positions; detachable parts removed		P
	Removal of lamps: protection against contact with live parts		P
	Use of test finger: no contact with live parts		P
8.1.2	Use of test pin: no contact with live parts		P
	Test pin is used at openings in earthed metal under certain conditions		N
8.1.3	Use of test probe: no contact with live parts of visible glowing heating elements		N
8.1.4	Accessible part not considered live if:		N
	- extra-low a.c. voltage: peak values not exceeding 42,4V		N
	- extra-low d.c. voltage: not exceeding 42,4V		N
	- or separated from live parts by protective impedance, d.c. current not exceeding 2 mA		N
	- or separated from live parts by protective impedance, a.c. peak value not exceeding 0,7 mA		N
	- for peak value 42,4V up to and including 450V capacitance not exceeding 0,1 μF		N
	- for peak value 450V up to and including 15 kV capacitance not exceeding 0,1 μF		N
8.1.5	Live parts protected at least by basic insulation before installation or assembly: checked by inspection and the test of 8.1.1:		N
	- built-in appliances		N
	- fixed appliances		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	- separate units		N
	compliance is checked by 8.1.1		N
8.2	Class II appliances and constructions adequately protected against accidental contact with basic insulation and metal parts separated from live parts with only basic insulation		N
	Only possible to touch parts separated from live parts by double or reinforced insulation		N
	Appliances with batteries replaceable by the user, basic insulation between live parts and the inner surface of the battery compartment adequate		N
	If appliance can be operated without batteries: double or reinforced insulation used		N
9.	Starting of motor-operated appliances		N
10	Power input and current		P
10.1	Power input at rated voltage and normal operating temperature not deviating from rated input by more than shown in table; measured power input (W); rated input (W); deviation .....:	The deviation of the power input not deviate from ±10%	P
10.2	Current at normal operating temperature not deviating from rated current by more than shown in table; measured current at rated voltage under normal operation (A); rated current (A); deviation :	The current deviation not deviate from ±10%	P
11	Heating		P
11.1	No excessive temperatures in normal use		P
(60335-2-29)	For appliances incorporating a swivel connection, compliance is also checked by the test of 11.101.	Clause of 11.101	P
11.2	Placing and mounting of appliance as described		P
(60335-2-29)	Appliances intended to be used on a stand or attached to a support are placed to give the most unfavourable results.		N
11.3	Temperature rises determined by thermocouples or resistance method	Temperature rises determined by thermocouples	P
11.4	Heating appliances operated under normal operation at 1,15 times rated power input	Heating appliances	P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
(60335-2-29)	If the temperature rise limits are exceeded in appliances incorporating motors, transformers or electronic circuits, and the power input is lower than the rated power input, the test is repeated with the appliance supplied at 1,06 times rated voltage.		N
11.5	Motor-operated appliances operated under normal operation at most unfavourable voltage between 0,94 and 1,06 times rated voltage		N
11.6(60335-2-29)	Combined appliances are operated as heating appliances.		N
11.7(60335-2-29)	Appliances without a timer are operated		P
	– for 30 min, for hand-held appliances;		P
	– in cycles of 30 s on and 5 s off until steady conditions are established, for hand dryers that are automatically controlled by the presence of the hands;		P
	– until steady conditions are established, for other appliances.		P
	Appliances incorporating a timer are operated in cycles until steady conditions are established. Each cycle consists of the maximum operating time of the timer followed by a rest period of 5 s.		P
11.8	Protective devices do not operate		P
	Sealing compound not flowing out		N
	Temperatures not exceeding values in table 3		P
(60335-2-29)	The temperature rise limits of motors, transformers and components of electronic circuits, including parts directly influenced by them, may be exceeded when the appliance is operated at 1,15 times rated power input.		N
11.101	Appliances incorporating a swivel connection are positioned with their major axis horizontal, the supply cord hanging vertically. A pull force of 1 N is applied to the supply cord.	A pull force of 1 N is applied to the supply cord	P

13	Leakage current	**P**

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
13.1	Leakage current not excessive and electric strength adequate		P
13.2	Leakage current measured by means of circuit described in Annex G		P
	Leakage current measurements	(see appended table)	P
13.3	Electric strength test of insulation	(see appended table)	P
	No breakdown during the test		P
15	Moisture resistance		P
15.1	Enclosure provides the degree of moisture protection according to classification of appliance		N
15.1.1	Appliance subjected to test as specified		N
	Withstand electric strength test specified in 16.3		N
	No trace of water on insulation which can result in a reduction of distances and clearances below values specified in 29.1		N
15.1.2	Hand-held appliance turned continuously through the most unfavourable positions during the test		N
	Built-in appliance installed according to the manufacturer's instruction		N
	Other appliances tested as specified		N
	parts to be removed by aid of tool , are not removed		N
15.2	Spillage of liquid does not affect the electrical insulation		N
	Overfilling test with additional amount of liquid (l) :		N
	Withstand electric strength test in 16.3		N
	No trace of water on insulation which can result in reduction of distances and clearances below values specified in 29.1		N
15.3	Humidity treatment for 48 h	Put into humidity treatment for 48 h	P
	Humidity is 93 ± 3%		P
	Withstanding the test of Cl. 16		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
16	Leakage current and electric strength		P
16.1	No excessive leakage current and adequate insulation and electric strength (tests 16.2 and 16.3)		P
16.2	Leakage current measurements	(see appended table)	P
16.3	Electric strength tests (values in table 5)	(see appended table)	P
17	Overload protection of transformers and associated circuits		N
	No excessive temperatures in transformer or associated circuits in event of short-circuits likely to occur in normal use		N
	Appliance supplied with 1,06 or 0,94 times rated voltage and the most unfavourable short-circuit or overload likely to occur in normal use applied		N
	Temperature rise of insulation of the conductors of safety extra-low voltage circuits not exceeding the relevant value specified in table 3 by more than 15 K		N
	Temperature of the winding not exceeding the value specified in table 6		N
	values are not applicable for fail-safe-transformers accord. 15.5) of IEC 61558- 1		N
18.	Endurance		N
19	Abnormal operation		P
19.1	The risk of fire or mechanical damage under abnormal or careless operation obviated		P
	Electronic circuits so designed and applied that a fault will not render the appliance unsafe		P
(60335-2-29)	Hairdryers are also subjected to the tests of 19.101 and 19.102.		N
19.2	Test of appliance with heating elements with restricted heat dissipation; test voltage (V): power input of 0,85 times rated power input .....:		P
(60335-2-29)	Restricted heat dissipation is obtained as follows:		-
	– motors are disconnected;		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	– hand-held hairdryers are placed on the floor of the test corner in any stable position likely to occur;		N
	– appliances intended to be filled with water are operated empty.		N
	Heaters for detachable curlers are placed on a piece of low-density glass-fibre insulation having a coefficient of thermal insulation of approximately 2,5 m2K/W.		N
19.3	Test of 19.2 repeated; test voltage (V): power input of 1,24 times rated power input .....:		P
19.4	Test conditions as in Cl. 11, the power input being 1,15 times rated power input, any control limiting the temperature during tests of Cl. 11 short-circuited		N
19.5	Test of 19.4 repeated on Class 0I and I appliances with tubular sheathed or embedded heating elements. No short-circuiting, but one end of the element connected to the elements sheath		N
	The test repeated with reversed polarity and the other end of the heating element connected to the sheath		N
19.6	Appliances with PTC heating elements tested as specified. Supplied at working voltage, establishing steady conditions, then the voltage increased in steps by 5% until 1,5 times at working voltage is reached or until the heating element ruptures		P
19.7	Stalling test by locking the rotor if the locked rotor torque is smaller than the full load torque or locking moving parts		N
	Locked rotor, motor capacitors open circuited or short-circuited, if required		N
	Appliances with timer or controller supplied with rated voltage for each of the tests, for a period equal to the maximum period allowed		N
	Test period at rated voltage (s or min) or until steady state conditions established .....:		P
	Winding temperatures not exceeding limiting temperature; type of appliance; insulation class; measured temperature (°C) .....:		N
	test is not repeated at capacitors of class P2 of IEC 60252		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
(60335-2-23)	The test is carried out for 5 min except for		P
	– hand-held appliances;		N
	– appliances that have to be kept switched on by hand;		N
	– appliances incorporating a timer.		P
19.8	Three-phase motors operated at rated voltage with one phase disconnected		N
19.9(60335-2-29)	A running overload test is carried out on appliances incorporating motors		N
19.10	Series motor operated at 1,3 times rated voltage for 1 min		N
	Parts not ejected from the appliance during test		N
(60335-2-29)	The test is carried out with the heating elements disconnected or switched off.		P
19.101	Hairdryers are operated as specified in Clause 11 except that the motor is supplied separately at its working voltage.		N
19.102	Portable hair dryers are operated under normal operation at 1,15 times rated power input.		N
19.11	Electronic circuits, compliance checked by evaluation of the fault conditions specified in 19.11.2 for all circuits or parts of circuits, unless they comply with the conditions specified in 19.11.1		P
19.11.1	Before applying the fault conditions a) to f) in 19.11.2, it is checked if circuits or parts of circuit meet both of the following conditions:		N
	- the electronic circuit is a low-power circuit, that is, the maximum power at low-power points does not exceed 15 W according to the tests specified		N
	- the protection against electric shock, fire hazard, mechanical hazard or dangerous malfunction in other parts of the appliance does not rely on the correct functioning of the electronic circuit		N
19.11.2	Fault conditions applied one at a time, the appliance operated under conditions specified in Cl. 11, but supplied at rated voltage, the duration of the tests as specified:		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	a) short-circuit of creepage distances and clearances between live parts of different potential, if these distances are less than the values specified in 29.1, unless the relevant part is adequately encapsulated	The creepage distances and clearance of different potential before fuse are not less than the values specified in 29.1.	N
	b) open circuit at the terminals of any component		P
	c) short-circuit of capacitors, unless they comply with IEC 384-14 or 14.2 of IEC 65		N
	d) short-circuit of any two terminals of an electronic component, other than integrated circuits. This fault condition is not applied between the circuits of an optocoupler		N
	e) failure of triacs in the diode mode		N
	f) failure of an integrated circuit. In this case the possible hazardous situations of the appliance are assessed to ensure that safety does not rely on the correct functioning of such a component		N
	During and after each test the following is checked:		P
	- the temperature rise of the windings do not exceed the values specified in table 6		P
	- the appliance complies with the conditions specified in 19.13		P
	- live parts not accessible to the test finger or test pin as specified in Cl. 8		P
	- any current flowing through protective impedance not exceeding the limits specified in 8.14		P
	If a conductor of a printed board becomes open circuited, the appliance is considered to have withstood the particular test, provided all three of the following conditions are met:		N
	- the material of the printed circuit board withstands the burning test of 20.1 of IEC 65		N
	- any loosened conductor does not reduce the creepage distances or clearances between live part and accessible metal parts		N
	- the appliance withstands the tests of 19.11.2 with open circuited conductor bridged		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
19.12	If the safety of the appliance for any of the fault conditions specified in 19.11.2 depends on the operation of a miniature fuse-link complying with IEC 127, the test is repeated, measuring the current flowing through the fuse-link; measured current (A); rated current of the fuse-link (A) .....:		N
19.13	During the tests the appliance does not emit flames, molten metal, poisonous or ignitable gas in hazardous amounts		P
	After the tests and when the appliance has cooled to approximately room temperature, the enclosure shall not have deformed to such an extent that compliance with clause 8 is impaired and the appliance shall comply with 20.2 if it can still be operated.		P
20	Stability and mechanical hazards		P
20.1	Adequate stability		N
	Tilting test through an angle of 10° (appliance placed on an inclined plane/horizontal plane); appliance does not overturn		N
	appliance is tilted 10 ° in most unvavourable direction		N
	Tilting test repeated on appliances with heating elements, angle of inclination increased to 15°		N
	Possible heating test in overturned position; temperature rise does not exceed values shown in table 7		N
20.2	Moving parts adequately arranged or enclosed as to provide protection against personal injury		P
	Protective enclosures, guards and similar parts are non-detachable	Enclosure fixed by screws	P
	Adequate mechanical strength and fixing of protective enclosures		P
	Self-resetting thermal cut-outs and overcurrent protective devices not causing a hazard, if unexpectedly reclosed		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Not possible to touch dangerous moving parts with test finger		P
21.	Mechanical strength		P
21.1	Appliance has adequate mechanical strength and is constructed as to withstand rough handling		P
	The appliance is rigidly supported and three blows, having an impact energy of 0.5 J, are applied to every point of the enclosure that is likely to be weak.	0.5J 3 times, After the test, the appliance no damage	P
(60335-2-29)	Hand-held appliances are also subjected to the test of 21.101		N
21.101(60335-2-29)	The appliance is placed on a horizontal surface positioned 700 mm above a rigidly		P
21.2	Accessible parts of solid insulation shall have sufficient strength to prevent penetration by sharp implements.		P
22	Construction		P
22.1	Appliance marked with the first numeral of the IP system: relevant requirements of IEC 529 are fulfilled		P
22.2	Stationary appliance: means to provide all-pole disconnection from the supply provided, the following means being available:		N
	- a supply cord fitted with a plug		N
	- a switch complying with 24.3		N
	- a statement in the instruction sheet that a disconnection incorporated in the fixed wiring is to be provided		N
	- an appliance coupler		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Single-phase Class I appliance with heating elements, intended to be permanently connected to fixed wiring, incorporating single-pole switches or single-pole protective devices for the disconnection of the heating element(s): the switches/devices being connected in the phase conductor		N
22.3	Appliance provided with pins: no undue strain on socket-outlets		P
	Applied torque not exceeding 0,25 Nm		P
22.4	Appliance for heating liquids and appliance causing undue vibration not provided with pins for insertion into socket-outlets		N
22.5	No risk of electric shock when touching the pins of the plug		P
22.6	Electrical insulation not affected by condensing water or leaking liquid		N
	Electrical insulation of Class II appliances not affected in case of a hose rupture or seal leak		N
22.7	Adequate safeguards against the risk of excessive pressure in appliances provided with steam-producing devices		N
22.8	Electrical connections not subject to pulling during cleaning of compartments to which access can be gained without the aid of a tool, and which are likely to be cleaned in normal use		N
22.9	Insulation, internal wiring, windings, commutators and slip rings not exposed to oil, grease or similar substances		P
	Adequate insulating properties of oil or grease to which insulation is exposed		P
22.10	Location or protection of reset buttons of non-self-resetting controls is so that accidental resetting is unlikely		N
22.11	Reliable fixing of non-detachable parts which provide the necessary degree of protection against electric shock, moisture or contact with moving parts	Enclosure fixed by screws	P
	Obvious locked position of snap-in devices used for fixing such parts		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	No deterioration of the fixing properties of snap-in devices used in parts which are likely to be removed during installation or servicing		P
	Tests	Pull & push, 50N	P
22.12	Handles, knobs etc. fixed in a reliable manner		N
	Fixing in wrong position of handles, knobs etc. indicating position of switches or similar components not possible		N
	Axial force 15 N applied to parts, the shape of which being so that an axial pull is unlikely to be applied		N
	Axial force 30 N applied to parts, the shape of which being so that an axial pull is likely to be applied		N
22.13	Unlikely that handles, when gripped as in normal use, make the operators hand touch parts having a temperature rise exceeding the value specified for handles which are held for short periods only		P
22.14	No ragged or sharp edges creating a hazard for the user in normal use, or during user maintenance		P
	No exposed pointed ends of self tapping screws etc., liable to be touched by the user in normal use or during user maintenance		P
22.15	Storage hooks and the like for flexible cords smooth and well rounded	No such parts	N
22.16	Automatic cord reels cause no undue abrasion or damage to the sheath of the flexible cord, no breakage of conductors strands, no undue wear of contacts	No such parts	N
	Cord reel tested with 6000 operations, as specified		N
	Electric strength test of 16.3, voltage of 1000V applied		N
22.17	Spacers not removable from the outside by hand or by means of a screwdriver or a spanner	No such parts	N
22.18	Current-carrying parts and other metal parts resistant to corrosion under normal conditions of use		P
22.19	Driving belts not used as electrical insulation	No such parts	N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
22.20	Direct contact between live parts and thermal insulation effectively prevented, unless material used is non-corrosive, non-hygroscopic and non-combustible		N
	compliance is checked and if necessary tested		N
22.21	Wood, cotton, silk, ordinary paper and fibrous or hygroscopic material not used as insulation, unless impregnated		P
	Remark: MgO and mineral ceramic fibres are not hygroscopic		N
22.22	Asbestos not used in the construction of the appliance		P
	No asbestos used		P
22.23	Oils containing polychlorinated biphenyl (PCB) not used		P
22.24	The heating element shall also be unlikely to come into contact with the skin or hair if it ruptures.		P
	In case of rupture, the heating conductor is unlikely to come in contact with earthed metal parts or accessible metal parts		P
(60335-2-29)	The heating element shall also be unlikely to come into contact with the skin or hair if it ruptures.		P
22.25	Sagging heating conductors cannot come into contact with accessible metal parts		P
22.26	The insulation between parts operating at safety extra-low voltage and other live parts complies with the requirements for double or reinforced insulation		N
22.27	Parts connected by protective impedance separated by double or reinforced insulation		N
22.28	Metal parts of Class II appliances conductively connected to gas pipes or in contact with water: separated from live parts by double or reinforced insulation		N
22.29	Class II appliances permanently connected to fixed wiring so constructed that the required degree of protection against electric shock is maintained after installation		N
22.30	Parts serving as supplementary or reinforced insulation fixed so that they cannot be removed without being seriously damaged, or		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	So constructed that they cannot be replaced in an incorrect position, and so that if they are omitted, the appliance is rendered inoperable or manifestly incomplete		P
22.31	Creepage distances and clearances over supplementary and reinforced insulation not reduced below values specified in 29.1 as a result of wear		P
	Creepage distances and clearances over supplementary or reinforced insulation not reduced to less than 50% of values specified in 29.1 if wires, screws etc. becomes loose		P
22.32	Supplementary and reinforced insulation designed or protected against deposition of dirt or dust		P
	Ceramic material not tightly sintered, similar material or beads alone not used as supplementary or reinforced insulation		N
	Supplementary insulation of natural or synthetic rubber resistant to ageing, or arranged and dimensioned so that creepage distances are not reduced below values specified in 29.1		N
	Oxygen bomb test at 70 °C for 96 h and 16 h at room temperature		P
(60335-2-29)	Supplementary insulation and reinforced insulation in class II curling irons shall be resistant to aging.		N
22.33	Conductive liquids which are or may become accessible in normal use are not in direct contact with live parts		N
	Conductive liquids not in direct contact to live parts (electrodes for heating)		N
	in class –II- constructions touchabl liquids not in contact to basic insulation or reinforced insulation		N
	in class –II- constructions liquids not in contact to reinforced insulation		N
22.34	Shafts of operating knobs, handles, levers etc. not live, unless the shaft is not accessible when the part is removed		P
22.35	Handles, levers and knobs, held or actuated in normal use, not becoming live in the event of an insulation fault		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Such parts being of metal, and their shafts or fixings are likely to become live in the event of an insulation fault, they are either adequately covered by insulation material, or their accessible parts are separated from their shafts or fixings by supplementary insulation		P
	This requirement does not apply to handles, levers and knobs on stationary appliances other than those of electrical components, provided they are either reliably connected to an earthing terminal or earthing contact, or separated from live parts by earthed metal		N
22.36	Handles continuously held in the hand in normal use are so constructed that when gripped as in normal use, the operators hand is not likely to touch metal parts, unless they are separated from live parts by double or reinforced insulation		N
(60335-2-29)	Contact with skin or hair in normal use shall be separated from live parts by double For class I appliances, other than hand dryers and face dryers, metal parts that could be in insulation or reinforced insulation and shall not be earthed.	Not such appliances	N
22.37	Capacitors in Class II appliances not connected to accessible metal parts, unless complying with 22.42		N
	Metal casings of capacitors in Class II appliances separated from accessible metal parts by supplementary insulation, unless complying with 22.42		N
22.38	Capacitors not connected between the contacts of a thermal cut-out		N
22.39	Lamp holders only used for the connection of lamps	No lampholder used	N
22.40	Motor-operated appliances and combined appliances, intended to be moved while in operation or which have accessible moving parts, are fitted with a switch to control the motor	Heating appliance	N
	The actuating member of this switch easily visible and accessible		N
(60335-2-29)	The switch in the off-position shall disconnect electronic circuits, unless compliance with Clause 19 does not depend on the operation of a self-resetting thermal cut-out.		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
22.101(60335-2-29)	Appliances having steam-producing or spray-producing devices shall be constructed so that there is no spillage or unintentional burst of steam or water that is likely to cause a hazard.	Not steam-producing or spray-producing device	N
22.102(60335-2-29)	Curling rollers of permanent-wave appliances having integral heating elements shall be supplied with safety extra-low voltage not exceeding 24V.		P
22.41	No mercury used		P
22.Z2	Fully halogenated chlorofluorocarbons (CFC's) not used		P
22.41	Mercury switches mounted according to the requirement		N
22.42	Protective impedance consisting of at least two separate components		P
	Values specified in 8.1.4 not exceeded if any one of the components is short-circuited or open circuited		P
22.43	Appliances adjustable for different voltages, accidental changing of the setting of the voltage unlikely to occur		N
22.44	Not designed as children toy	Forbidden for children	P
23	Internal wiring		P
23.1	Wireways smooth and free from sharp edges		P
	Wires protected against contact with burrs, cooling fins etc.		P
	Wire holes in metal well rounded or provided with bushings		P
	Wiring effectively prevented from coming into contact with moving parts		P
23.2	Beads etc. on live wires cannot change their position, and are not resting on sharp edges or corners		N
	Beads inside flexible metal conduits contained within an insulating sleeve		N
23.3	Electrical connections and internal conductors movable relatively to each other not exposed to undue stress		P
	Flexible metallic tubes not causing damage to insulation of conductors		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Open-coil springs not used		N
	Adequate insulating lining provided inside a coiled spring, the turns of which touch one another		P
	No damage after 10 000 flexings		P
	Electric strength test, 1000 V between live parts and metal parts		P
(60335-2-29)	The number of flexings for conductors that are only flexed when the appliance is stored is 5 000.		P
23.4	Bare internal wiring sufficiently rigid and fixed		N
23.5	The basic insulation of internal wiring withstanding the electrical stress likely to occur in normal use	For internal wires	P
	No breakdown when a voltage of 2000V is applied for 15 min between the conductor and metal foil wrapped around the insulation	2000V, 15 Min	P
23.6	Sleeving used as supplementary insulation on internal wiring retained in position by positive means		P
23.7	Only the colour combination green/yellow used for earthing conductors	No green-yellow wires used.	N
23.8	Aluminium wires not used for internal wiring		P
23.9	No lead-tin soldering of stranded conductors where they are subject to contact pressure, unless		N
	Clamping means so constructed that there is no risk of bad contact due to cold flow of the solder		P
24	Components		P
24.1	Components comply with safety requirements in relevant IEC standards	See appended table	P
24.1.1	Capacitors likely to be subjected to the supply mains voltage and used for radio interference suppression or voltage dividing, comply with Annex Q		P
	Small lampholders: compliance with requirements for E10 lampholders		N
	Safety isolating transformer wich not comply with IEC 61558-2-6 shall complxy with annex R		N
	Appliance couplers for IPx0 appliances: compliance with IEC 320		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Automatic controls: compliance with IEC 730, unless tested with the appliance		N
	Other appliance couplers: compliance with IEC 309		N
	Switches shall comply with IEC 61058 –1 or tested in the appliance	Approved switch, comply with IEC 61058-1, 10000 cycles	P
24.1.2	Automatic controls complying with IEC 730: additional tests according to this standard and 11.3.5 to 11.3.8 and Cl. 17 of IEC 730 as type 1 controls, the cycles of operation being:		N
	- thermostats: 10 000		N
	- temperature limiters: 1000		N
	- self-resetting thermal cut-outs: 300		N
	- non-self-resetting thermal cut-outs: 30		N
	energy regulators 10.000		N
	timers 3000		N
24.1.3	Switches ,which not comply IEC 61058 are tested accord. annex S		N
	10.000 cycles of operation		N
	Switches for no-load-operation and operable only with the aid of a tool, are not subjected to the tests of Cl. 17 of IEC 1058-1		N
	Switches for no- load-operation		N
(60335-2-29)	Switches incorporated in hand dryers are subjected to 50 000 cycles of operation.		N
24.1.4	Components marked with their operating characteristics are used in the appliance in accordance with these markings		P
	Components which have to comply with other standards are tested separately, according to the relevant standard		P
	Components used within the limits of its marking, tested in accordance with conditions occurring in the appliance		P
	Components not marked, or not used in accordance with its marking, or no IEC standard exists, tested under the conditions occurring in the appliance		P
	Components not mentioned in table 3 tested as part of the appliance		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
24.1.5	Voltage across capacitors in series with a motor winding does not exceed 1,1 times rated voltage, when the appliance is supplied at 1,1 times rated voltage under minimum load		P
	Capacitors permanent connected to motor windings and have to comlpy 30.2.3 shall be of class P1 or P2 of IEC 60252		N
	List of components		N
24.2(60335-2-29)	Helmet-type hairdryers and permanent-wave appliances may incorporate a switch in a flexible cord.	Not such device	N
	No thermal cut-outs which can be reset by soldering		N
	Remark:solder with melting point < 250 °C allowed		P
24.3	Switch intended for all-pole disconnection of stationary appliances is directly connected to the supply terminals, having a contact separation of at least 3 mm in each pole		N
24.4	Plugs and socket-outlets for heating elements and extra-low voltage circuits, not interchangeable with plugs, and		P
	Socket-outlets or with connectors and appliance inlets complying with IEC 83 or IEC 320, respectively add IEC 60906-1		N
24.5	Plugs and socket-outlets etc. for interconnection cords, not interchangeable with plugs and socket-outlets or connectors and appliance inlets complying with IEC 83 or IEC 320, respectively, if direct supply from the mains could give rise to a hazard add IEC 60906-1		N
24.6	Motors connected to the supply mains and having inadequate basic insulation for the rated voltage of the appliance, comply with the requirements of Annex F		N
25	Supply connection and external flexible cords		P
25.1	Appliance not intended for permanent connection to fixed wiring, means for connection to the supply:		P
	- supply cord fitted with a plug	Supply cord with plug	P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	- an appliance inlet having at least the same degree of protection against moisture as required for the appliance		N
	- pins for insertion into socket-outlets		N
25.2	Appliance not provided with more than one means of connection to the supply		P
	Stationary appliance for multiple supply may be provided with more than one means of connection, provided electric strength test of 1250 V for 1 min between each means of connection causes no breakdown	Not stationary appliance	N
25.3	Connection of supply wires for appliance intended to be permanently connected to fixed wiring possible after the appliance has been fixed to its support	Not intended to be connected to fixed wiring.	N
	Appliance provided with a set of terminals for the connection of cables or fixed wiring, cross-sectional areas specified in 26.2		N
	Appliance provided with a set of terminals allowing the connection of a flexible cord		N
	Appliance provided with a set of terminals and cable entries, conduit entries, knock-outs or glands, allowing connection of appropriate type of cable or conduit		N
25.4	Cable and conduit entries, rated current of appliance not exceeding 16 A, dimensions according to table 8		P
	Introduction of conduit or cable does not affect the protection against electric shock or reduce creepage distances and clearances below values specified in 29.1		P
25.5	Method for assemble supply cord with the appliance:		P
	- type X attachment		N
	- type Y attachment		P
	- type Z attachment, if allowed in part 2		N
	Type X attachment: specially prepared cord		N
	Type X attachment not used for flat twin tinsel cord		N
25.6	Plugs fitted with only one flexible cord		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Supply cords of single-phase portable appliances having a rated current not exceeding 16 A, provided with a plug complying with the following Standard Sheets of IEC 83:		N
	- for Class I appliances: Standard Sheet C2b, C3b or C4		N
	- for Class II appliances: Standard Sheet C5 or C6		N
25.7	Appliance supply cord not lighter than:		-
	- braided cord (245 IEC 51)		N
	- ordinary tough rubber sheathed cord (245 IEC 53)		N
	- ordinary polychloroprene sheathed flexible cord (245 IEC 57)		N
	- flat twin tinsel cord (227 IEC 41)		N
	- light polyvinyl chloride sheathed cord (227 IEC 52), appliance not exceeding 3 kg		P
	- ordinary polyvinyl chloride sheathed cord (227 IEC 53), appliance exceeding 3 kg		N
	Temperature rise of external metal parts exceeding 75 K, PVC cord not used		N
	PVC cord used: appliance so constructed that the supply cord is not likely to touch external metal parts in normal use		P
	PVC supply cord appropriate for higher temperatures, type Y or type Z attachment used		N
	Power supply cord with high flexibility not lighter than:		N
	- rubber sheated cord (60245 IEC 86) flexibility		N
	- sheathed cord of net-PVC (60245 IEC 88) flexibility		N
	- sheathed cord with rubber insulation and supply insulation of net-PVC (60245 IEC 87) flexibility		N
25.8	Nominal cross-sectional area of supply cords according to table 9; rated current (A); cross-sectional area (mm^2) .....:	3x0.75mm^2	P
25.9	Supply cord not in contact with sharp points or edges		P
25.10	Green/yellow core for earthing purposes in Class I appliance		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
25.11	Conductors of supply cords not consolidated by lead-tin soldering where they are subject to contact pressure, unless	Solder terminal not subject to contact pressure.	N
	Clamping means so constructed that there is no risk of bad contacts due to cold flow of the solder		N
25.12	Moulding the cord to part of the enclosure does not damage the insulation of the supply cord		N
25.13	Inlet opening provided with a bushing, or is so constructed, that there is no risk of damage to the supply cord when introduced	Bushing mould on the cord	P
25.13.1	Inlet bushing so shaped as to prevent damage to the supply cord		N
	Inlet bushing not detachable		N
25.13.2	At inlet openings, the insulation between the conductor of a supply cord and the enclosure of the appliance is consisting of the insulation of the conductor, and in addition:		N
	- for Class 0 appliances: at least one separate insulation		N
	- for other appliances: at least two separate insulations		N
	Only one separate insulation is required if the enclosure at the inlet opening is of insulating material		N
	The separate insulation consists of:		P
	- the sheath of a supply cord at least equivalent to that of a cord complying with IEC 227 or 245		P
	- a lining or bushing of insulating material complying with the requirements of 29.2 for supplementary insulation		P
25.14	Supply cords adequately protected against excessive flexing		P
	Flexing test; applied force (N); number of flexings .....:	10N, 10000 flexings	P
	The test does not result in:		P
	- short-circuit between the conductors		P
	- breakage of more than 10% of the strands of any conductor	<10% of the strands of conductor	P
	- separation of the conductor from its terminal		P
	- loosening of any cord guard		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	- damage, within the meaning of the standard, to the cord or the cord guard		P
	- broken strands piercing the insulation and becoming accessible		P
25.15	Conductors of the supply cord relieved from strain, twisting and abrasion by use of cord anchorages		P
	The cord cannot be pushed into the appliance to such an extent that the cord or internal parts of the appliance can be damaged		P
	Pull and torque test of supply cord, values shown in table 10: pull (N); torque (Nm) (not on automatic cord reel) .....:	30N, 0,1Nm	P
	Max. 2 mm displacement of the cord, and conductors not moved more than 1 mm in the terminals	0,7 mm	P
	Creepage distances and clearances not reduced below values specified in 29.1		P
25.16	Cord anchorages for type X attachments so constructed and located that:		N
	- replacement of the cord is easily possible		N
	- it is clear how the relief from strain and the prevention of twisting are obtained		N
	- they are suitable for different types of cord		N
	- cord cannot touch the clamping screws of cord anchorage if these screws are accessible, unless separated from		N
	- accessible metal parts by supplementary insulation		N
	- the cord is not clamped by a metal screw which bears directly on the cord		N
	- at least one part of the cord anchorage securely fixed to the appliance, unless part of a specially prepared cord		N
	- screws which have to be operated when replacing the cord do not fix any other component, if applicable		N
	- if labyrinths can be bypassed the test of 25.15 is nevertheless withstood		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	- for Class 0, 0I and I appliances: they are of insulating material or are provided with an insulating lining, unless a failure of the insulation of the cord does not make accessible metal parts live		N
	- for Class II appliances: they are of insulating material, or if of metal, they are insulated from accessible metal parts by supplementary insulation		N
25.17	Adequate cord anchorages for type Y and Z attachment	Type Y attachment	P
25.18	Cord anchorages only accessible with the aid of a tool, or		P
	So constructed that the cord only can be fitted with the aid of a tool		P
25.19	Type X attachment, glands not used as cord anchorage in portable appliances		N
	Tying the cord into a knot or tying the cord with string not used		N
25.20	Conductors of the supply cord for type Y and Z attachment adequately additionally insulated	Type Y attachment	P
25.21	Space for supply cable for fixed wiring or supply cord for type X attachment constructed to permit checking of conductors with respect to correct positioning and connection before fitting any cover, no risk of damage, no contact with accessible metal parts if a conductor becomes loose, etc.		N
	For portable appliances, the uninsulated end of a conductor prevented from any contact with accessible metal parts, unless the end of the cord is such that the conductors are unlikely to slip free		N
25.22	Appliance inlet:		N
	- live parts not accessible during insertion or removal	No appliance inlet	N
	- connector can be inserted without difficulty		N
	- the appliance is not supported by the connector		N
	- is not for cold conditions if temperature rise of external metal parts exceeds 75 K, unless the supply cord is not likely to touch such metal parts		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
25.23	Interconnection cords comply with the requirements for the supply cord, except as specified		N
	If necessary, electric strength test of 16.3		N
25.24	Interconnection cords not detachable without the aid of a tool		N
26	Terminals for external conductors		P
26.1.1	Appliances with type X attachment and appliances for connection to fixed wiring provided with terminals in which connection is made by means of screws, nuts or equally effective devices		N
	Screws and nuts serve only to clamp supply conductors, except		N
	Internal conductors, if so arranged that they are unlikely to be displaced when fitting the supply conductors		N
26.1.2	For type X attachment soldered connections used, the conductor so positioned or fixed that reliance is not placed on soldering alone		N
	Soldering alone used, barriers provided, creepage distances and clearances satisfactory if the conductor becomes free		N
	For type Y and Z attachment: soldered, welded, crimped and similar connections used	Supply conductors soldered to PCB	P
	For Class II appliances: the conductor so positioned or fixed that reliance is not placed on soldering, welding or crimping alone	Cable tie used near the terminals	N
	For Class II appliances: soldering, welding or crimping alone used, barriers provided, creepage distances and clearances satisfactory if the conductor becomes free		N
26.2	Terminals for type X attachment and for connection to fixed wiring suitable for connection of conductors with required cross-sectional area according to table 11; rated current (A); nominal cross-sectional area (mm²) .....:		N
	Terminals only suitable for a specially prepared cord		N
26.3	Terminals for the supply cord suitable for their purpose		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Terminals with screw clamping and screwless terminals not used for flat twin tinsel cords, unless conductors ends fitted with a device suitable for screw terminals		N
	Pull test of 5 N to the connection		N
26.4	Terminals for type X attachment and those for connection to fixed wiring so fixed that when tightening or loosening the clamping means:		N
	- the terminal does not loosen		N
	- internal wiring is not subjected to stress		N
	- creepage distances and clearances are not reduced below the values in 29.1		N
26.5	Terminals for type X attachment and for connection to fixed wiring so constructed that the conductor is clamped between metal surfaces with sufficient contact pressure and without damaging the conductor		N
26.6	Terminals for type X attachment, no special preparation of conductors required, and so constructed and placed that conductors prevented from slipping out, except those with a specially prepared cord and those for connection to fixed wiring		N
26.7	Terminals of the pillar type constructed and located as specified		N
26.8	Terminals for the connection to fixed wiring located close to each other, including the earthing terminal		N
26.9	Terminals for type X attachment accessible after removal of a cover or part of the enclosure		N
26.10	terminals shall only be accessible after removal of a non-detachable part		P
26.11	Terminals for type X attachment so located or shielded that if a wire of a stranded conductor escapes, no risk of accidental connection between live parts and accessible metal parts,		N
	and for Class II construction, between live parts and metal parts separated from accessible metal parts by supplementary insulation only		N
	Stranded conductor test, 8 mm insulation removed		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
27	Provision for earthing		P
27.1	Accessible metal parts of Class 0I and I appliances, permanently and reliably connected to an earthing terminal		N
	Earthing terminals not connected to neutral terminal		P
	Class 0, II and III appliance have no provision for earthing		N
27.2	Screwless terminals comply with IEC 998-2-2		P
	Terminals used for the connection of external equipotential bonding conductors allow connection of conductors of 2,5 to 6 mm², and		P
	do not provide earthing continuity between different parts of the appliance		P
	Conductors cannot be loosened without the aid of a tool		N
	Clamping means adequately secured against accidental loosening		N
27.4	No risk of corrosion resulting from contact between metal of earthing terminal and other metal		P
	Adequate resistance to corrosion of coated or uncoated parts providing earthing continuity, other than parts of a metal frame or enclosure		P
	Parts of steel providing earthing continuity provided at the essential areas with an electroplated coating, thickness at least 5 μm		N
	Adequate protection against rusting of parts of coated or uncoated steel, only intended to provide or transmit contact pressure		N
	In case of aluminium alloys precautions taken to avoid risk of corrosion		N
27.5	Low resistance of connection between earthing terminal and earthed metal parts		P
	Resistance not exceeding 0,1 Ω at the specified low-resistance test	<0.1Ω	P
27.6	In hand-held appliances printed conductors of printed circuit boards not used to provide earthing continuity		N
	In other appliances at least two tracks are used with independent soldering points, and		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	the appliance complies with the requirements of 27.5 for each circuit, and		N
	the material of the printed board complies with IEC 249-2-4 or IEC 249-2-5		N
28	Screws and connections		P
28.1	Fixings and electrical connections withstand mechanical stresses ...." and connections providing earthing continuity".		P
	Screws not of soft metal liable to creep, such as zinc or aluminium		P
	Diameter of screws of insulating material min. 3 mm		P
	Screws of insulating material not used for any electrical connection ...." and connections providing earthing continuity".		N
	Screws transmitting electrical contact only screwing into metal		P
	Screws not of insulating material if their replacement by a metal screw can impair supplementary or reinforced insulation		P
	Type X attachment, screws to be removed for replacement of supply cord, or for users maintenance, not of insulating material if their replacement by a metal screw can impair basic insulation		N
	Screws and nuts transmitting contact pressure subjected to torque test as specified, applying torque as shown in table 12	Torque test: 0,5Nm for screws	P
	The test is not carried out on screws and nuts transmitting contact pressure for earthing continuity provided at least two screws or nuts are used		N
28.2	Contact pressure not transmitted through insulating material liable to shrink or distort, unless shrinkage or distortion compensated ...." and connections providing earthing continuity".		P
	requirement doesn`t apply to circuits < 0,5 A		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
28.3	Space-threaded (sheet metal) screws only used for the connection “electrical connections“if they clamp these parts directly in contact with each other		N
	Thread-cutting (self-tapping) screws not used for electrical connection of current-carrying parts, unless generating a full form standard machine screw thread		N
	Thread-cutting (self-tapping) screws not used if they are likely to be operated by the user or installer unless the thread is formed by a swaging action		N
	Thread-cutting and space-threaded screws used provide earthing continuity:		N
	- it is not necessary to disturb the connection in normal use		N
	- two screws used for each connection		N
28.4	Screws and nuts making mechanical connection between different parts of the appliance, and also making electrical connection or providing earthing continuity secured against loosening		N
	Rivets for current-carrying connections subject to torsion secured against loosening		N
29	Creepage distances, clearances and distances through insulation		P
29.1	Creepage distances and clearances not less than specified in table 13		P
	Resonant voltage between the point where a winding and a capacitor are connected together and metal parts separated from live parts by basic insulation only, creepage distances and clearances not less than the values specified for the value of the voltage produced by the resonance		N
	Values increased by 4 mm in case of reinforced insulation when resonance voltage		N
29.2	Distances through insulation not less than 1,0 mm for supplementary insulation, and 2,0 mm for reinforced insulation		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
29.2.1	Supplementary insulation applied in thin sheet form, other than mica or similar scaly material, consists of at least two layers, each of the layers withstands the electric strength test of 16.3 for supplementary insulation		N
	Reinforced insulation applied in thin sheet form, other than mica or similar scaly material, consists of at least three layers, and any two of the layers together withstand the electric strength test of 16.3 for reinforced insulation		N
29.2.2	Supplementary or reinforced insulation inaccessible and does not exceed the maximum permissible temperature values		P
	Supplementary or reinforced insulation, after conditioning as specified, withstands the electric strength test as specified in 16.3, both at the oven temperature and room temperature		P
30	Resistance to heat, fire and tracking		P
30.1	See Annex H		P
	Relevant external parts of non-metallic material		P
	Parts supporting live parts and parts providing supplementary or reinforced insulation sufficiently resistant to heat		P
	Ball-pressure test with a force of 20 N, diameter of impression not exceeding 2 mm		P
	External parts: at 75 °C		P
	Parts supporting live parts: at 125 °C		P
	Parts providing supplementary or reinforced insulation: temperature (°C) .....:		P
30.2	Relevant parts of non-metallic material adequately resistant to ignition and spread of fire		P
30.2.1	Possible burning test of relevant parts according to Annex J		N
	Glow-wire test of Annex K made at temperature 550 °C	Enclosure: 650 °C	P
30.2.2	Appliances operated while attended, parts of insulating material supporting connections carrying a current exceeding 0,5 A in normal operation, subjected to the glow-wire test of Annex K at 750 °C	PCB: 650 °C	P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
30.2.4	Parts of non-metallic material within a distance of 50 mm from parts not withstanding the tests of 30.2.2 or 30.2.3, subjected to the needle-flame test of Annex M		N
30.3	Relevant insulating material have adequate resistance to tracking		N
	Tracking test at 175V according to Annex N		N
	Tracking test at 250V according to Annex N		N
	No hazard other than fire, tracking test at 175V according to Annex N, and in addition needle-flame test of surrounding parts according to Annex M		N
	Possible needle-flame test of non-metallic material		N
31	Resistance to rusting		P
	Relevant ferrous parts adequately protected against rusting		P
32	Radiation, toxicity and similar hazards		P
	Appliance does not emit harmful radiation		P
	Appliance does not present a toxic or similar hazard		P
A	Annex A, normative references		P
	The annex contains a list of standards which are referred to, and thus become part of, this standard		P
B	Annex B, testing of appliances powered by rechargeable batteries		N
B.2	Definitions		N
B.2.2.9	Appliances operated under the following conditions:		N
	- the appliance supplied by its fully charged battery is operated as specified in part 2		N
	- the appliance is charged, the battery being initially discharged to such an extent that the appliance cannot operate		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	- if possible, the appliance is supplied from the supply mains through its battery charger, the battery being initially discharged to such an extent that the appliance cannot operate. The appliance is operated as specified in part 2		N
B.2.7.2	If a part has to be removed in order to discard the battery before scrapping the appliance, this part is not considered to be detachable even if the instructions state that it is to be removed		N
B.4	General conditions for the tests		N
B.4.101	Unless otherwise specified, appliances supplied from the supply mains are tested as specified for motor-operated appliances		N
B.7	Marking and instructions		N
B.7.1	Battery compartment for batteries intended to be replaced by the user, marked with battery voltage and polarity of the terminals		N
B.7.12	The instructions for appliances incorporating batteries intended to be replaced by the user, include required information		N
	Details given about how to remove batteries containing materials hazardous to the environment		N
	Materials which are hazardous to the environment are mercury, cadmium or lead		N
B.7.15	Markings placed on the part connected to the supply mains		N
B.8	Protection against access to live parts		N
B.8.2	Basic insulation between live parts and parts accessible during and after removal of the battery insulation during replacement of batteries		N
B.11	Heating		N
B.11.7	Charging time for the battery		N
B.19	Abnormal operation		N
B.19.101	Charging time at rated voltage		N
B.19.102	Short-circuiting of the battery, fully charged, for appliances having batteries which can be removed without the aid of a tool		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
B.19.103	Appliances having batteries replaceable by the user, supplied at rated voltage under normal operation with the battery removed or in any position allowed by the construction		N
B.21	Mechanical strength		N
B.21.101	Appliances having pins for insertion into socket-outlets, checked according to procedure 2 of IEC 68-2-32		N
	Mass of part not exceeding 250 g, 100 falls		N
	Mass of part exceeding 250 g, 50 falls		N
B.22	Construction		N
B.22.3	Appliances having pins for insertion into socket-outlets are tested as fully assembled as possible		N
B.25	Supply connection and external flexible cords		N
B.25.13.2	The requirement is not applicable to interconnection cords subjected to safety extra-low voltage		N
B.30	Resistance to heat, fire and tracking		N
B.30.2	For parts connected to the supply mains during the charging period, 30.2.3 applies		N
	For other parts, 30.2.2 applies		N
C	Annex C, ageing test on motors		N
	Test carried out when doubt with regard to the classification of the insulating system of a motor winding		N
D	Annex D, alternative requirements for protected motor units		N
	Void		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
E	Annex E, measurement of creepage distances and clearances		P
	Methods of measuring creepage distances and clearances, specified in 29.1, indicated in 10 different cases		P
F	Annex F motors not isolated from the supply mains and having basic insulation not designed for the rated voltage of the appliance		N
	Motors having a working voltage not exceeding 42 V, not being isolated from the supply mains, and having basic insulation not designed for the rated voltage of the appliance are tested according to this annex		N
	All clauses of this standard apply, unless otherwise specified in this annex		N
F.8	Protection against accessibility to live parts		N
F.11.8	Temperature rise of the body of the motor, where in contact with insulating material, not exceeding values in table 3 for the relevant insulating material		N
F.16	Leakage current and electric strength		N
F.19	Abnormal operation		N
F.19.101	Appliance operated at rated voltage with each of the following defects:		N
	- short-circuit of the terminals of the motor, including any capacitor incorporated in the motor circuit		N
	-short cut on diodes of rectifier		N
	- open circuit of the supply to the motor		N
	- open circuit of any shunt resistor during operation of the motor		N
F.22	Construction		N
F.22.101	Class I appliance incorporating a motor supplied by a rectifier circuit, the d.c. circuit being insulated from accessible parts of the appliance by double or reinforced insulation		N
G	Annex G, circuit for measuring leakage currents		P
	Meassuring circuit for leakage current		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
H	Annex H, selection and sequence of the tests of clause 30		P
J	AnnexJ, burning test		N
	The burning test is made in accordance with IEC 707, and method FH is used		N
	Category FH3 applies, the maximum burning rate being 40 mm/min		N
K	Annex K, glow-wire test		P
	The glow-wire test is made in accordance with IEC 695-2-1 (clause numbers between parentheses refer to IEC 695-2-1)		P
(4)	Description of test apparatus: the last paragraph before the note is replaced		P
(5)	Severities: the duration of application of the tip of the glow-wire to the specimen being (30 ± 1) s	Glow-wire test: 550 °C for 30 s for enclosure Glow-wire test: 650 °C for 30 s for PCB,	P
(10)	Observations and measurements: item c) does not apply		P
L	Annex L, bad-connection test with heaters		P
	The bad-connection test with heaters is made in accordance with IEC 695-2-3 (clause numbers between parentheses refer to IEC 695-2-3)		P
(3)	General description of the test: additions concerning crimped connections		P
(4)	Description of test apparatus: replacements of some of the test specifications and the first paragraph of the note		P
(6)	Severities: the duration of application of the test power being (30 ± 1) min		P
(8)	Test procedure: 8.6 replaced		P
(11)	Information to be given in the relevant specification: item h), the first dashed paragraph, does not apply		P
M	Annex M, needle-flame test		N
	The needle-flame test is made in accordance with IEC 695-2-2 (clause numbers between parentheses refer to IEC 695-2-2)		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
(4)	Description of the apparatus: the sixth paragraph is replaced		N
(5)	Severities: the duration of application of the test flame is (30 ± 1) s		N
(8)	Test procedure: some changes in the test specifications		N
(10)	Evaluation of the test results: addition in the test specification		N
N	Annex N, proof tracking test		N
	The proof tracking test is made in accordance with IEC 112 (clause numbers between parentheses refer to IEC 112)		N
(3)	Test specimen: the last sentence of the first paragraph does not apply		N
(5)	Test apparatus: some changes in the subclauses		N
(6)	Procedure: adjustments of the test specifications		N
P	Annex P, severity of duty conditions of insulating material with respect to the risk of tracking		P
	Recognition of different duty conditions with respect to the risk of tracking	Normal duty condition	P
Q	Annex Q, capacitors		P
	Testing of capacitors		P
	modifications of IEC 60384 – 14		P
R	Safety isolating transformers		N
R 7.1	Transformer shall markede with:		N
	name , trade mark		N
	model or type		N
R.17	fail-safe transformers shall comply with 1.5 of IEC 61558-1		N
R.22	19.1 , 19.2 of IEC 61558-2-6 are applicable		N
R.29	distances accord. 2a,2b,3 of table 13 of 61558-1		N
S	Switches		N
	Switches are tested additional clauses accord. IEC 61058 – 1		N

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	20 no load operation		N
8	marking of trade mark or manufacturer , type		N
13	this clause is applicable		-
15.3	applicable for full disconnection and micro-disconnection		N
17	compliance is checked at three separate samples		N
	temperature rise at terminals not more as 30 k above values measured at clause 11		N
20	appicable for full disconnection and micro-disconnection switches		N
	CDand CL for live parts as stated in table 18		N

ZA	Annex ZA, special national conditions		—
7.12	Denmark: requirements regarding marking tag of power supply cord and connecting of earthing wire		—
19.5	Norway: the test is also applicable to appliances intended to be permanently connected to fixed wiring		—
19.11.2	Austria: requirements regarding appliances having circuits which under fault conditions may cause earth-leakage currents having a d.c. component exceeding 5 Ma and exceeding 20% of the total earth-leakage		—
22.2	France, norway: The second paragraph of this subclause dealing with single-phase Class I appliances with heating elements is not applicable due to the supply system		—
25.6	Belgium, france, greece, united kingdom: plugs according to Standard Sheet C2b not allowed		—
	Austria, germany, finland, iceland, ireland, italy, luxembourg, netherlands, norway, portugal, spain, sweden, switzerland, united kingdom: plugs according to Standard C3b not allowed		—
	Denmark: Supply cords of single-phase portable appliances having a rated current not exceeding 10 A provided with a plug according to the following:		—

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
	Class I appliances: Section 107-2-DI Standard Sheet DK2-1a		—
	For appliances covered by a Part 2 of EN 60 335, also plugs in accordance with IEC 83, Standard Sheet C2b, C3b or C4 are allowed		—
	Class II appliances: IEC 83, Standard Sheet C5 or C6		—
	Stationary single-phase appliances, having a rated current not exceeding 10 A, and provided with a plug, the plug is in accordance with the requirements above		—
	Multi-phase appliances and single-phase appliances having a rated current exceeding 10 A, and provided with a plug, the plug is in accordance with the requirements below:		—
	Class I appliances: Section 107-2-D1, Standard Sheet DK6-1a/EN 60 309-2, Standard Sheet 2-II, 2-IV		—
	Class II appliances: Section 107-2-D1, Standard Sheet DK6-1a/2-II, 2-IV		—
	Ireland: plug is in accordance with Standard Sheets B1 (15A), B2 and C2b		—
	SPAIN: Appliances having a rated current not exceeding 6 A, provided with a plug complying with UNE 20 315:		-
	for Class I appliances: Figure 7C		—
	for Class II appliances: Figure 15A		—
	Class I appliances having a rated current not exceeding 16 A, provided with a plug complying with Standard UNE 20 315 Figure 7B		—
	Switzerland: supply cords of portable household and similar electrical appliances, rated current not exceeding 10 A, provided with a plug complying with SEV 1011 or IEC 884-1 and one of the following dimension sheets:		—
	SEV 6532-2:1991 plug type 15 3P+N +PE 250/400V, 10 A		—
	SEV 6532-2:1991 plug type 11 L+N 250V, 10 A		—
	SEV 6532-2:1991 plug type 12 L+N+PE 250V, 10 A		—
	United kingdom: plug according to Standard Sheet B2 or C5 used (refer to Annex ZB)		—

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**
25.8	Ireland, united kingdom: replacement of figures (rated current/cross-sectional area) in the table		—
ZB	Annex ZB, a-deviations		—
3	Switzerland: information about batteries		—
7.1	Italy: the voltage is 220V / 380V		—
	SPAIN: the voltages are 127V/22V and 220V/380V		—
7.12	Ireland: information about required label attached to the supply cord, concerning the colour code of the wires		—
22.22	Germany: the amount of asbestos in the mass containing asbestos not exceeding 0,1%		—
	Finland: certain types of asbestos not used		—
25.6	United kingdom: regulations concerning plugs to be fitted to domestic appliances		—
ZF	Annex ZF, informative		P
	IEC and CENELEC code designations for flexible cords		P

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**

11.8	TABLE: temperature rise measurements		P
	Heating appliances	Supplied power input: 1,15 times rated power input	—
Thermal test (Normal operation)			—

Temperature rise dT of part/at:	Measured temperature dT(k)	Limit dT (K)
Supply cord	6.3	50
PCB	49.5	85
Switch	5.2	30
Internal wire	18.6	105

Note:
1. The ambient temperature during this test was 24.1-24.6°C;
2. The value of dT is calculated under the condition the temperature of ambient considered as 25℃

13.2	TABLE: leakage current measurements at operating temperature		P
	heating appliances:	at 1,15 times rated input (W)	—
	motor-operated and combined appliances: at 1,06 times rated voltage (V) .....:	N/A	—

leakage current I between:	I (mA)	required I (mA)
L/N – enclosure	0.007	0,75

13.3	TABLE: electric strength measurements at operating temperature		P

test voltage applied between:	test voltage (V)	breakdown
L-N	1000	No
L/N – enclosure	1000	No

16.2	TABLE: leakage current measurements		P
	at 1,06 times rated voltage (V) .....:	233V	—

leakage current I between:	I (mA)	required I (mA)
L/N – enclosure	0.006	0,75

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**

16.3	TABLE: electric strength measurements	P

test voltage applied between:	test voltage (V)	breakdown
L-N	2000	No
L/N – enclosure	2000	No

	ANNEX 1: components			P
object/part No.	type/model	technical data	manufacturer/ trademark	mark(s) of conformity
Power plug	SF-230	12V	SUN FAI INDUSTRIAL CO.,LTD	ASTA 1026
Power cord	H05VV-F	3×0.75 mm ^{2}	Various	VDE
Internal wire	FEP-101	180DegC 22AWG	GUANZHOU FENGTAI ELECTRONIC CO.LTD	VDE:136964
PCB	01V0	V-0; 130°C	WENZHOU GALAXY ELECTRONICS CO LTD	E157634
Switch button	ABS PA-757	12V	CHI MEI CORPORATION	VDE

IEC 60335-2-29-2016			
Clause	**Requirement – Test**	**Result - Remark**	**Verdict**

EMF TEST FOR DL.D908

Requirement & Testing	**Remark**	**Verdict**
Measuring method 4.2.4.1: (time domain evaluation)		P
Measuring method 4.2.4.2: (line spectrum evaluation)		N
Measuring method 4.2.4.3: (simplified test method)		N
Measuring distance (cm):	30	P
Sensor location:	Around	P
Operation conditions:	Highest temperature setting	P
Magnetic flux density (maximum)(μT):		P
If the measured value exceeds the reference level, the value is weighted with coupling factor:		N
Weighted result, W=		N
Coupling factor:	0.17	P
Test Result	24.3%	P

Photo of sample

SOLAR CHARGERS

SOLAR CHARGERS BRAND:SUNAFRICA MODEL:2045 MADE IN CHINA YIWU MOJIA TRADING CO.,LTD

End of Report